Tiny object detection with SNIP, SNIPER and AutoFocus

by LAB Team

CONTENT

SNIP
Scale Normalization for Image Pyramids
01

SNIPER
Scale Normalization for Image Pyramids with Efficient Resampling
02

AutoFocus
FocusPixels - FocusChips - FocusStacking
03

SNIP

Scale Normalization for Image Pyramids

Image classification at multiple scales

Naive Multi-scale Inference:

- CNN-B model
  - Step 1: downsample ImageNet dataset from 224 to 48, 64, 80, 96, 128
  - Step 2: upsample above images back to 224
  - Step 3: train CNN (ResNet101) that pretrained with 224 images

=> The more different in resolution between train set and test set, the lower accuracy

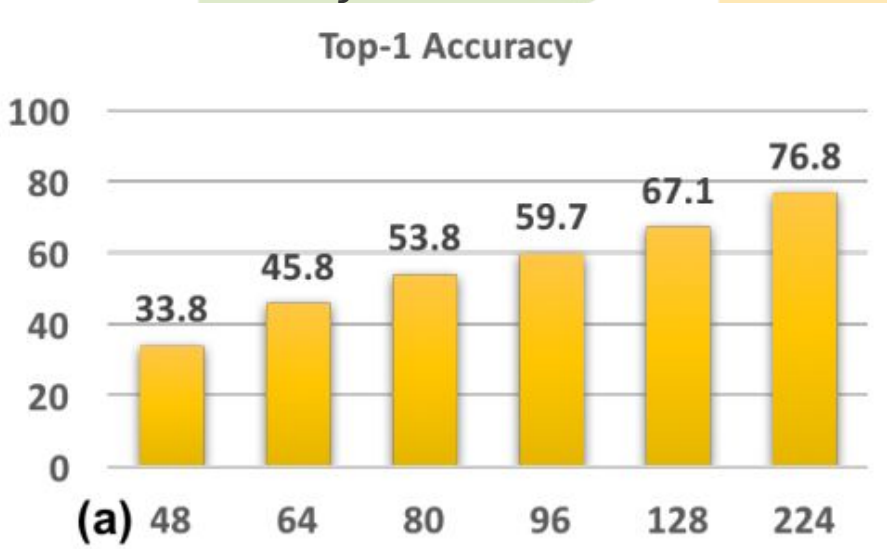


Image classification at multiple scales

Resolution Specific Classifiers:

- CNN-B: First conv layer is 7x7, stride = 2, follow by max pooling stride = 2x2
- CNN-S:
  - First conv layer is 3x3, stride = 1 (for 48x48 images)
  - First conv layer is 5x5, stride = 2 (for 96x96 images)
  - Data augmentation: random crop, color augmentation (in first 70 epochs)

Image classification at multiple scales

Fine-tuning High-resolution Classifiers:

- CNN-B-FT:
  - Fine-tuning CNN-B on up-sampled low resolution images

Image classification at multiple scales

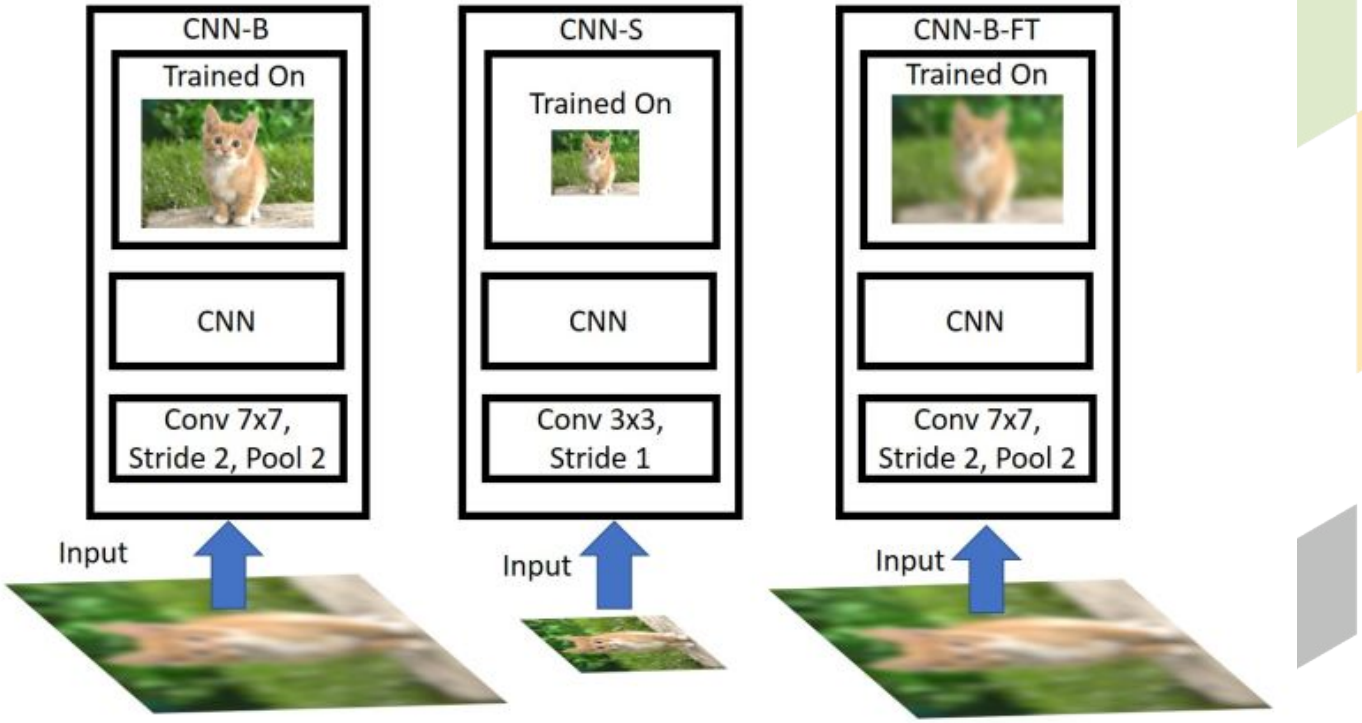
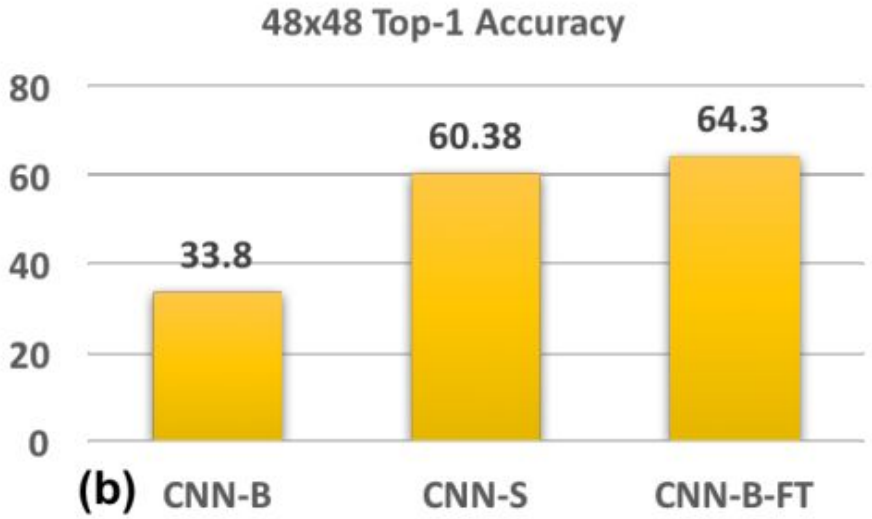
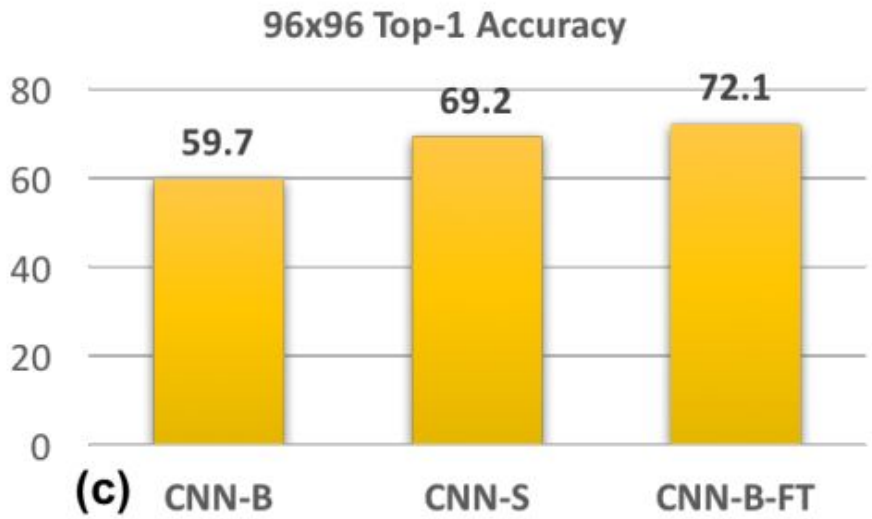


Image classification at multiple scales





Data Variation vs Correct Scale

- Train with different settings
- Evaluate on 1400x2000 images in COCO dataset (small object is smaller than 32x32 pixels)

Data Variation vs Correct Scale

Train with different resolution:

- 800_{all}: Train with image 800x1400, backprop all object instances
- 1400_{all}: Train with 1400x2000, backprop all object instances

=> 1400_{all} > 800_{all} (19.9 > 19.6 in mAP)
=> Improvement is not significant
=> with 1400_{all}, scaling up small objects improves, but medium-to-large objects become too large and degrade model performance

Data Variation vs Correct Scale

Scale specific detector:

- $1400_{<80px}$: Train with 1400x2000, backprop only object instances that smaller than 80x80 pixels

=> 800_{all} > $1400_{<80px}$ (19.6 > 16.4 in mAP)
=> Significantly worse
=> with $1400_{<80px}$, lots of data of medium-to-large instance have been ignored (30% of the total object instances)

Data Variation vs Correct Scale

Multi scale training:

- MST: Train with images that randomly sample from multiple resolutions

=> 800_{all} ~ MST (19.6 > 19.4 in mAP)
=> Almost similar
=> with MST, instances are observed in multiple resolutions, but model cannot learn extreme small or large objects

Data Variation vs Correct Scale

Scale Normalization for Image Pyramids:

- SNIP:
  - Modified version of MST
  - Backprop only instances whose resolution is close to pretrained dataset (or fall in the desired scale range)
  - Ignore the remainder

=> Reduce the domain-shift and work in detecting small objects

$1400_{<80px}$	800_{all}	1400_{all}	MST	SNIP
16.4	19.6	19.9	19.5	21.4

SNIPER

Scale Normalization for Image Pyramids with Efficient Resampling

Problem of SNIP

- Process image multiple times
- Process all pixels of image

=> SLOW

=> SNIPER performs as **well** as **SNIP** while **reduce** number of **pixels** processed by **a factor of 3** on **COCO** dataset

Chip Generation

- Generate chips C^i at scale s_i
- Resize image
- Place KxK pixel chip at equal intervals of d pixels

=> List of chips

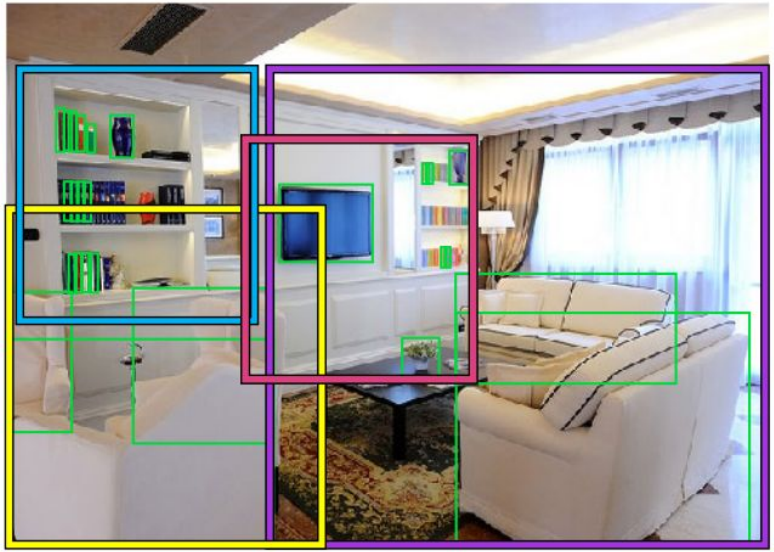
Positive Chip Selection

- Propose desired area for each scale $R^i = [r^i_{min}, r^i_{max}]$

=> List of valid bbox for each scale G_i

- Select chip that completely cover a bbox

=> List of positive chips C^i_{pos}



Negative Chip Selection

- Train RPN for a couple of epochs (no negative chips used)
- Generate proposals over the entire training set
- Assume: no proposals generated in portion = no object in portion
- Remove all the proposals covered in C^i_{pos}
- Select chip that cover at least M proposals in R^i

=> List of negative chips C^i_{neg}

Positive and Negative Chips

AUTO FOCUS

FocusPixels - FocusChips - Focus Stacking

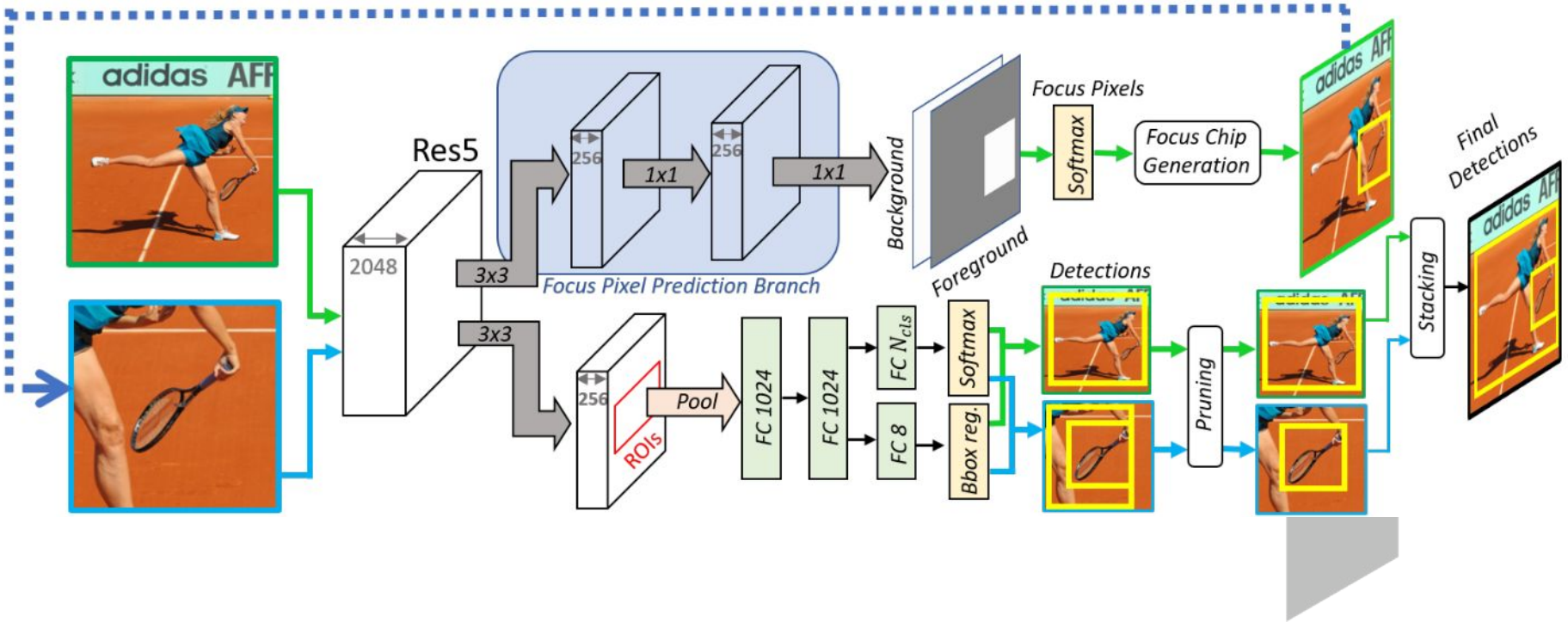
Problem of SNIPER

- Process the entire high resolution image during inference

=> SLOW
=> AutoFocus proposes a framework that can **automatically predict** these chips for **small objects** at a **coarser scale**

Moreover,
AutoFocus proposes an algorithm which correctly **merges detections** from chips at **multiple scales**

AutoFocus


Res5
2048
3x3
3x3
256
1x1
256
1x1
Focus Pixel Prediction Branch
Background
Foreground
Focus Pixels
Softmax
Focus Chip Generation
Final Detections
Detections
Stacking
Pruning
256
Pool
ROIs
FC 1024
FC 1024
FC N_{cls}
FC 8
Softmax
Bbox reg.

FocusPixels

- Author **labels pixel** from **features map Res5** of ResNet101
- A **pixel** is a **FocusPixel** if it has any **overlap** with a **small object** (range between **5x5** and **64x64**) => positives
- A **pixel** is invalid if it has **overlap** with **object** (range smaller than 5x5 or between **64x64** and **90x90**)
- All other pixels are negatives

$$l = \begin{cases} 1, & IoU(GT, l) > 0, a < \sqrt{GTArea} < b \\ -1, & IoU(GT, l) > 0, \sqrt{GTArea} < a \\ -1, & IoU(GT, l) > 0, b < \sqrt{GTArea} < c \\ 0, & \text{otherwise} \end{cases}$$

FocusChips

Algorithm 1: FocusChip Generator

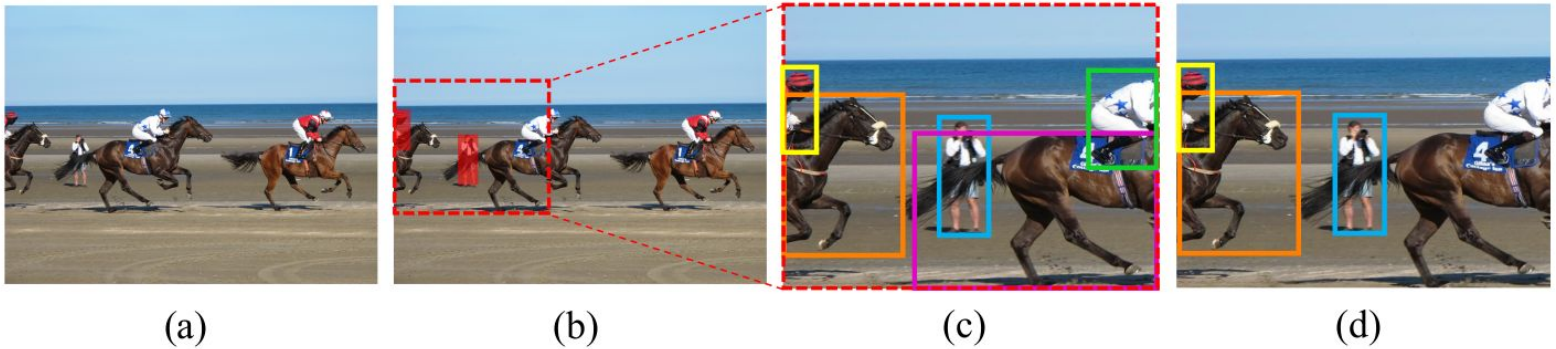
Input : Predictions for feature map $\mathcal{P}$, threshold t, dilation constant d, minimum size of chip k

Output: Chips $\mathcal{C}$

1 Transform $\mathcal{P}$ into a binary map using the threshold t
2 Dilate $\mathcal{P}$ with a $d \times d$ filter
3 Obtain a set of connected components $\mathcal{S}$ from $\mathcal{P}$
4 Generate enclosing chips $\mathcal{C}$ of size $> k$ for each component in $\mathcal{S}$
5 Merge chips $\mathcal{C}$ if they overlap
6 **return** *Chips* $\mathcal{C}$

FocusStacking

- **Cascaded multi-scale inference** causes generating some **false positives**
- Step 2 in FocusChips Generator ensures **no *interesting* object** at the next scale is at the **boundaries** of the **chip**
- These ***interesting* object** should be in **another chip**
- If **detections** are observed at the **boundaries**, **discard** them
  - Detail of discarding policy is based on detection boundary, chip boundary and image boundary
- **Non-Maximum Suppression** is applied to aggregate the detections




PIXTA
VIETNAM